

Why Study Primate Diet to Understand Human Diet?



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Note: Make sure you have read Chapter 1 in Gremillion prior to working through this unit of the module.

As Gremillion begins her book in the reading you completed for today, to understand the diet of people in the past (and the present!), we must look to our evolutionary origins, which help scientists understand how and when we acquired our unique dietary practices. Although she begins her discussion with the advent of multicellular life and heterotrophy (consuming other organisms to gain energy), most relevant to our course is a discussion of how understanding our evolutionary history as **primates** has constrained our food choices over time.

The most important aspect of the evolutionary history of primates is that they evolved in tropical forests and adapted well to eat resources available in that environment. Through application of the **comparative approach**, it is possible to use what we know about primates most closely related to the human lineage (i.e., other apes, especially chimpanzees and bonobos) to make inferences about the diets of early **hominins**. This is one method for making inferences about the diet of our immediate ancestors, of whom we have little direct evidence save fossils—and, in the case of some of our potential ancestors, very few fossils.

Primates are an order of mammals that evolved between 85 million and 55 million years ago. Shared characteristics include large brains, a highly developed sense of vision (including full color vision), flexible shoulders, and dexterous hands (with opposable thumbs, in most species). Many primates are social, living in groups.

The **comparative approach** is used by biological anthropologists and paleoanthropologists to understand the relationship between anatomical features and behaviors by drawing on the shared history of evolutionary adaptations to environmental constraints and opportunities between the species under investigation. For example, to understand the diet of an extinct species of hominin, of whom we only have a single fossil mandible (jawbone), we may compare the teeth of that hominin to the teeth of different primate species living today. If we find a good match, we can see what foods that primate species eats and use that parallel to make inferences about what the hominin had adapted to eat.

Hominin is the term used to reference **bipedal** primates: humans and our extinct ancestors of the genus *Homo* and the earlier genus *Australopithecus*, as well as other smaller or taxonomically uncertain branches of extinct bipedal primates. Technically, the term hominin comes from the tribe Hominini, the evolutionary clade that is usually defined to include both humans (our species and many extinct lineages) and the genus *Pan*, of which two species (chimpanzees and bonobos) exist today. [Note: This is different from Gremillion's description in Figure 1.1 on page 10, but our definitions of "hominin" are the same.] In the archaeological literature and in this class, we use the term hominin to refer exclusively to human species of the genus *Homo*, both living and extinct, and the extinct hominins of our lineage since the split from the genus *Pan*. This split occurred at some point between 5 million and 13 million years ago. Extended genetic exchange among these closely related lineages makes it difficult to trace

the timing of this divergence using genetic methods alone, and fossils have been inconclusive to date.

Bipedal primates are those who habitually walk on two legs. While bipedality is not unique to our lineage (kangaroos, for example, are bipedal, as are birds!), only hominins are bipedal among the primates. Chimpanzees and gorillas, among other primates, can walk on two feet, but return to walking on all fours regularly.